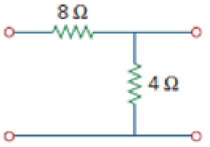
Problem

Find the z parameters of the two-port network in Fig. 19.9.

Figure 19.9



Step-by-step solution

Step 1 of 8

Refer to Figure 19.9 in the text book.

Connect the voltage source $\mathbf{V}_1$ as shown in Figure 1. Identify the currents and voltages in the circuit.

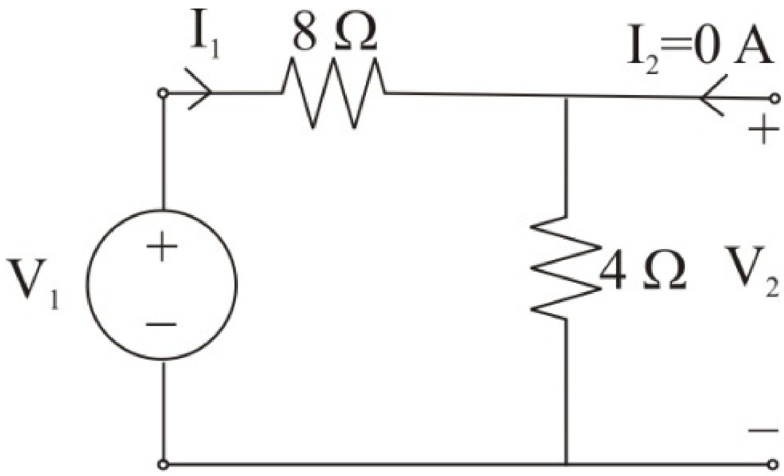


Figure 1

Step 2 of 8

Consider the expression for the voltage $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(8+4)\\ &= \mathbf{I}_1(12)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_1(12)$.

Consider the expression for the voltage $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{V}_2 &= \mathbf{I}_1(4)\\ &= \mathbf{I}_1(4)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_1(4)$.

Step 3 of 8

Determine the expression for the open circuit input impedance $\mathbf{z}_{11}$.

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1}$$

Substitute $\mathbf{I}_1(12)$ for $\mathbf{V}_1$.

$$\mathbf{z}_{11} = \frac{\mathbf{I}_1(12)}{\mathbf{I}_1}$$
$$= 12\ \Omega$$

Therefore, the expression for the open circuit input impedance $\mathbf{z}_{11}$ is $\boxed{12\ \Omega}$.

Step 4 of 8

Determine the value of the open circuit transfer impedance $\mathbf{z}_{21}$ from port-2 to port-1.

$$\mathbf{z}_{21} = \frac{\mathbf{V}_2}{\mathbf{I}_1}$$

Substitute $\mathbf{I}_1(4)$ for $\mathbf{V}_2$.

$$\mathbf{z}_{21} = \frac{\mathbf{I}_1(4)}{\mathbf{I}_1}$$
$$= 4\ \Omega$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{21}$ from port-2 to port-1 is $\boxed{4\ \Omega}$.

Step 5 of 8

Connect the voltage source $\mathbf{V}_2$ as shown in Figure 2. Identify the currents and voltages in the circuit.

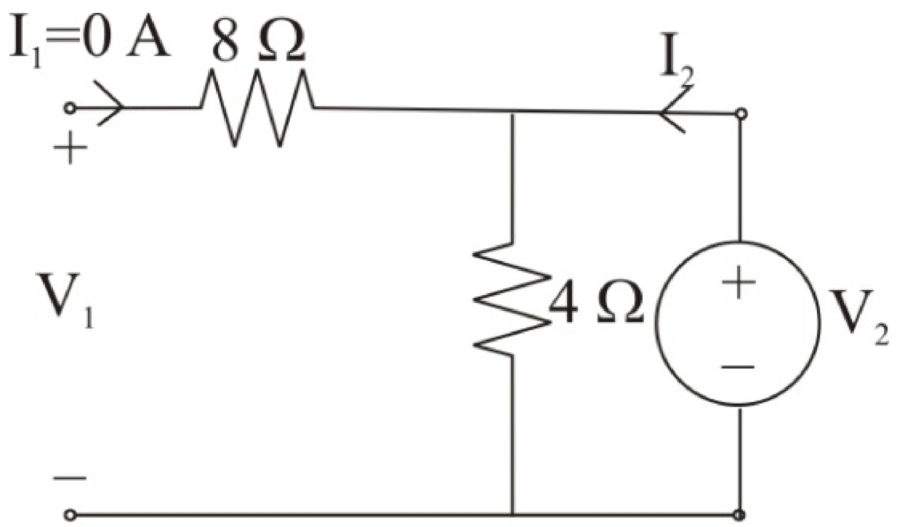


Figure 2

Step 6 of 8

Consider the expression for the voltage $\mathbf{V}_2$.

$$\mathbf{V}_2 = \mathbf{I}_2(4)$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_2(4)$.

Consider the expression for the voltage $\mathbf{V}_1$.

$$\mathbf{V}_1 = \mathbf{I}_2(4)$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_2(4)$.

Step 7 of 8

Determine the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2.

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(4)$ for $\mathbf{V}_1$.

$$\begin{aligned} \mathbf{z}_{12} &= \frac{\mathbf{I}_2(4)}{\mathbf{I}_2} \\ &= 4\ \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2 is $\boxed{4\ \Omega}$.

Step 8 of 8

Determine the expression for the open circuit output impedance $\mathbf{z}_{22}$.

$$\mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(4)$ for $\mathbf{V}_2$.

$$\begin{aligned} \mathbf{z}_{22} &= \frac{\mathbf{I}_2(4)}{\mathbf{I}_2} \\ &= 4\ \Omega \end{aligned}$$

Therefore, the expression for the open circuit output impedance $\mathbf{z}_{22}$ is $\boxed{4\ \Omega}$.